

Pre-Earnings Valuation and Event Risk

A Theoretical Framework for Fair Value, Implied Movement, and Scenario Uncertainty

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Abstract

This paper develops a theoretical framework for analyzing expected price movement around scheduled earnings events. The framework separates four objects that are often collapsed into one: fair value, the market-implied scale of the event move, the probability of discrete reaction states, and the extent to which a valuation gap can be incorporated into price over the event horizon.

The paper makes four principal contributions. First, it treats the at-the-money straddle as a risk-neutral expectation of an absolute move rather than as the size of a tail outcome. Second, it shows that a neutral state defined as an interval absorbs part of that absolute move, producing a closed-form leakage correction. Third, it derives the expected-value and risk expressions for both symmetric and asymmetric three-state reaction models. In the asymmetric model, the sign of expected value changes when the ratio of downside to upside move magnitudes equals the ratio of bullish to bearish probabilities. Fourth, it separates fair value from event price through an explicit convergence coefficient and derives a sampling-error floor for the directional probability tilt.

The paper is a theoretical and methodological specification. It does not present fitted parameters, a backtest, or evidence that a deployable forecasting system produces economic profits. Its central conclusion is therefore about identification: an apparent pre-earnings edge is not determined by a valuation gap alone. It depends jointly on event magnitude, directional probabilities, downside asymmetry, convergence, and estimation uncertainty.

This paper does not produce a security-specific expected-value forecast. Nor should it be treated as a model for producing alpha, a trading signal, or a deployable investment recommendation. The equations define theoretical identities and identify the inputs and empirical evidence that a future forecasting study would need; they do not supply those inputs or establish that an alpha exists.

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1 Introduction

Pre-earnings valuation models commonly combine a fair-value estimate with an expected move and then report a single expected return. That compression hides several distinct questions. A valuation model concerns the level at which a security might be fairly priced. An options market supplies a price for dispersion over a specified horizon. A scenario model assigns probabilities to possible reactions. A short-horizon price response need not equal the change in fair value. Finally, all of these quantities are estimated with error.

This paper develops a common language for those questions. It is deliberately more cautious than a trading rule. The object is a coherent theoretical model whose assumptions, identities, and failure modes are visible before any empirical estimation is attempted.

1.1 Research question

Let P_0 be the price immediately before the event and let P_1 be the price at the end of a fixed event window. The central question is:

Under what assumptions can a pre-earnings fair-value estimate imply a statistically distinguishable expected event return, once market-implied dispersion and asymmetric reaction risk are accounted for?

The answer is not a single formula. It is a sequence of distinctions. The framework first constructs a valuation gap, then describes event-scale movement, then assigns scenario probabilities, and finally states how the valuation gap can enter the event-price return. The layers can be estimated separately, but they must not be silently added together when they describe the same price movement.

1.2 Theoretical contribution

The paper contributes the following results.

1. **Event-scale identification.** An at-the-money straddle prices a risk-neutral mean absolute move, not a one-standard-deviation move and not the magnitude of a tail state. The three-state magnitude must therefore be matched to a probability-weighted absolute-move moment.
2. **Neutral-band correction.** When the neutral state is an interval, the interval contributes to the mean absolute move. Under a normal reference distribution, the fraction is $\ell_{\mathcal{N}}(p_N^{\text{ref}}) = 1 - \exp(-z^2/2)$, where $z = \Phi^{-1}((1 + p_N^{\text{ref}})/2)$.
3. **Asymmetric reaction result.** If downside and upside magnitudes differ, the expected-value sign condition is $p_U = \rho p_B$, where $\rho = m_B/m_U$. Equal tail probabilities are not sufficient for zero expected value once skew is admitted.
4. **Identification and precision.** A valuation gap reaches the event price through a convergence coefficient κ , while the directional edge is driven by a probability tilt whose sampling error must be compared with the tilt at the event counts available for the intended application.

1.3 Scope and non-claims

This is a theoretical preprint, not an empirical finance study. It does not claim that any parameter has been estimated from a completed event panel, that the probability model is calibrated out of sample, or that a position-sizing rule is deployable. Numerical values are explicitly labelled as identities, assumptions, or illustrations. The paper does not disclose production architecture, implementation details, private data workflows, or proprietary forecasting components.

The framework is useful even before estimation because it identifies what an empirical study would have to measure. A theoretical result that an edge changes sign with an unidentified parameter is itself a result about what cannot be inferred from the available inputs.

1.4 Organization

Section 2 defines the horizons, state space, and notation. Section 3 describes the fair-value layer. Section 4 derives the event-scale correction. Section 5 formulates scenario probabilities and the market-implied scale benchmark. Section 6 gives the fair-value-to-price bridge. Section 7 combines the layers without double counting and derives the principal closed forms. Section 8 studies uncertainty. Section 9 states empirical implications and limitations. Appendices collect secondary decision material and a fully labelled symbolic illustration.

2 Setup, Horizons, and Notation

2.1 Two horizons

An earnings model contains at least two horizons:

H_{event} = a prespecified earnings-event window, H_{value} = the horizon over which fair value is assessed.

A discounted valuation over H_{value} and an options-implied move over H_{event} are different objects. They may be related only by an explicit price-adjustment model. In particular, a multi-period valuation gap must not be treated as though it were a one-day expected return.

2.2 Price, value, and returns

Let P_0 and P_1 denote the pre-event and post-event market prices. Let V_0 and V_1 denote the corresponding model-implied fair values. Define

$$R_i^{\text{val}} = \frac{V_i - P_0}{P_0}, \quad R_i^{\text{px}} = \frac{P_i - P_0}{P_0},$$

where the subscript i denotes a reaction state. The superscripts distinguish the valuation return from the realized price return; they are not interchangeable.

2.3 State space

The event reaction is represented by an ordered state variable $Y \in \{B, N, U\}$: bearish, neutral, and bullish. Let

$$\mathbb{P}(Y = i) = p_i, \quad p_i \geq 0, \quad p_B + p_N + p_U = 1.$$

The three-state representation is a deliberate low-dimensional modeling choice. It does not claim that the true return distribution has only three points or that its tails are thin. A fixed threshold $a > 0$ defines the states before the outcomes are observed:

$$Y = \begin{cases} B, & R^{\text{px}} < -a, \\ N, & |R^{\text{px}}| \leq a, \\ U, & R^{\text{px}} > a. \end{cases}$$

The threshold is part of the model definition and is not identified by the theory. Choosing it after observing which partition produces a desirable expected value would make the result a tuning exercise rather than a theoretical prediction.

2.4 Notation

Symbol	Meaning	Definition or restriction
P_0, P_1	pre- and post-event market price	
V_0, V_1	pre- and post-event fair value	model-implied
R_i^{val}	value-gap return in state i	$(V_i - P_0)/P_0$
R_i^{px}	realized event-price return in state i	$(P_i - P_0)/P_0$
Y	reaction state	$Y \in \{B, N, U\}$
p_B, p_N, p_U	state probabilities	sum to one
a	neutral-state threshold	fixed before outcomes
m_{IV}	straddle-implied absolute move	approximately $\mathbb{E}^{\mathbb{Q}} R^{\text{px}} $
M	physical tail absolute-move budget	defined in Section 4
m_B, m_U	downside and upside state magnitudes	positive
ρ	downside-to-upside magnitude ratio	m_B/m_U
κ	value-to-price convergence coefficient	free estimand in the core model
EV	expected event return	statistical expected value only
γ	relative risk-aversion coefficient	utility parameter
Θ	enterprise value	never denoted by EV

2.5 Identification status of inputs

The equations below do not assign numerical values to the threshold a , the forecast horizon, the physical adjustment h , the tail-asymmetry ratio ρ , the convergence coefficient κ , the probability vector, the regularization weight η , or the utility and cost parameters. They are inputs, restrictions, or estimands for a later empirical design. The state count and the choice of a reference distribution are modeling choices rather than empirical facts. The unit scale in $\text{Logistic}(0, 1)$ is an identification normalization, and constants such as $\sqrt{2/\pi}$ are derived only under the explicitly stated normal reference model. Algebraic indices, probability normalization, and indicator values are notation, not calibrations.

3 The Fair-Value Layer

3.1 Valuation as a range, not a false point estimate

Different valuation methods often share earnings, growth, discount-rate, and capital-structure inputs. Their errors are therefore correlated, so mechanically averaging them does not create inde-

pendent information. A conservative summary is a range:

$$\boxed{V_0 \in [V_{\min}, V_{\max}], \quad V_{\min} = \min_j V_j, \quad V_{\max} = \max_j V_j.}$$

When a single reference value is needed for exposition, let $\hat{V}_0$ be the value selected by the stated valuation protocol and let ς_V be its uncertainty or dispersion summary. Neither choice is identified by the equations:

$$\hat{V}_0 \in [V_{\min}, V_{\max}], \quad \varsigma_V \geq 0.$$

Here ς_V is a protocol-defined uncertainty summary, not automatically a sampling standard error. A point estimate becomes a meaningful empirical object only when the valuation methods, their dependence, and their out-of-sample performance are specified.

3.2 Forward earnings valuation

For a k -period forecast path,

$$EPS_{t+k} = EPS_t \prod_{j=1}^k (1 + g_{E,j}), \quad V_{PE} = \frac{EPS_{t+k} PE_{t+k}}{(1+r)^k}.$$

The justified multiple should respect the curvature of a growth model. One optional structural anchor is the Gordon-style expression

$$PE^* = \frac{1-b}{r-g_E}, \quad g_E < r,$$

so a residual log-linear specification can be written as

$$\ln PE_f = \ln \left(\frac{1-b}{r-g_E} \right) + \alpha_1 ROIC + \alpha_2 M - \alpha_3 \sigma - \alpha_4 C + \varepsilon.$$

The anchor prevents a negative multiple and makes clear which terms are theoretical and which would require estimation.

3.3 Discounted cash flow

For completeness, an enterprise-value DCF is

$$\Theta_{DCF} = \sum_{t=1}^T \frac{FCF_t}{(1+WACC)^t} + \frac{TV_T}{(1+WACC)^T}, \quad TV_T = \frac{FCF_T(1+g_T)}{WACC-g_T},$$

and the implied equity value per share is

$$V_{DCF} = \frac{\Theta_{DCF} + Cash - Debt}{n_{sh}}, \quad g_T < WACC.$$

Terminal value is often the dominant term. The framework therefore treats the width of the valuation range as part of the result rather than hiding it inside a precise-looking discounted cash-flow output.

3.4 Valuation gap

The pre-event value gap is

$$D_V = \frac{V_0 - P_0}{P_0}.$$

Positive D_V means that the model's fair value exceeds the current price. If $\hat{V}_0$ and ς_V are used,

$$D_V \pm \frac{\varsigma_V}{P_0}$$

is more honest than a point estimate. A gap smaller than the valuation dispersion is not evidence of detectable mispricing.

3.5 Persistent fundamental regimes

If a valuation path depends on a latent growth regime G_t , Bayesian updating belongs to that regime rather than to the already-realized event outcome. Let $G_t \in \{G_B, G_N, G_U\}$. Then

$$\mathbb{P}(G_t \mid D_t) = \frac{\mathbb{P}(D_t \mid G_t)\mathbb{P}(G_t)}{\sum_j \mathbb{P}(D_t \mid G_j)\mathbb{P}(G_j)},$$

and persistence can be represented by

$$A_{ij} = \mathbb{P}(G_{t+1} = j \mid G_t = i), \quad \mathbb{P}(G_{t+1} = j) = \sum_i \mathbb{P}(G_t = i \mid D_t) A_{ij}.$$

This is a structural option for the valuation layer. It does not provide an empirical estimate of the transition matrix.

4 The Event-Scale Layer

4.1 What an at-the-money straddle measures

For an at-the-money straddle with strike close to the relevant forward price, use the shorthand that any required discounting, dividend, and spot-forward conversions have already been made, and define the observed premium on the event-window return scale as

$$m_{IV} \equiv \frac{C_{ATM} + P_{ATM}}{P_0} \approx \mathbb{E}^{\mathbb{Q}}[R^{\text{px}}].$$

The approximation suppresses the discounting, dividend, spot-forward, and strike adjustments needed to express the straddle premium as a return moment. Those adjustments must be made explicitly in an empirical implementation. It also does not identify the physical distribution. Under a centered short-horizon normal reference model,

$$m_{IV} \approx \sqrt{\frac{2}{\pi}} \sigma \sqrt{T}$$

Here σ is the reference volatility scale per unit time and T is the event-horizon length measured in the same time units. The normal expression is a derived identity conditional on that reference distribution; it is not a general property of the straddle. The risk-neutral absolute move is not automatically a physical expected move. Let $h > 0$ denote an explicit physical-to-risk-neutral scale adjustment. The value and even the direction of that adjustment are not identified by the theory and no numerical value is assigned to it here.

4.2 Tail magnitude in a symmetric three-state model

If $R^{\text{px}} \in \{-m, 0, +m\}$, then

$$\mathbb{E}|R^{\text{px}}| = m(p_B + p_U).$$

If the neutral state were a point mass at zero and $p_B + p_U > 0$, consistency with the adjusted absolute-move budget hm_{IV} would require

$$m = \frac{hm_{IV}}{p_B + p_U}.$$

This is already different from setting $m = m_{IV}$: that shortcut ignores both tail concentration and the unidentified physical-to-risk-neutral scale adjustment, so it is not generally justified.

4.3 Neutral-band leakage under a normal reference

The neutral state is an interval in the underlying return distribution, even though the three-state payoff approximation represents it by one value. To derive a reference correction, suppose a centered normal return $X \sim \mathcal{N}(0, s_{\mathcal{N}}^2)$ is used and the neutral band is $|X| \leq zs_{\mathcal{N}}$, so the threshold in Section 2 is $a = zs_{\mathcal{N}}$. Its reference probability is $p_N^{\text{ref}} = 2\Phi(z) - 1$, and

$$\mathbb{E}|X| = 2s_{\mathcal{N}}\varphi(0),$$

while the absolute move inside the neutral band is

$$\mathbb{E}[|X|\mathbf{1}\{|X| \leq zs_{\mathcal{N}}\}] = 2s_{\mathcal{N}}(\varphi(0) - \varphi(z)).$$

The fraction of the absolute-move budget absorbed by the neutral state is

$$\ell_{\mathcal{N}}(p_N^{\text{ref}}) = 1 - \frac{\varphi(z)}{\varphi(0)} = 1 - e^{-z^2/2}, \quad z = \Phi^{-1}\left(\frac{1 + p_N^{\text{ref}}}{2}\right).$$

If this normal reference is also used to construct the physical tail budget, define

$$M = hm_{IV}(1 - \ell_{\mathcal{N}}(p_N^{\text{ref}})).$$

If a conditional model supplies a physical p_N , that probability must remain separate from p_N^{ref} ; replacing the normal-reference probability with a conditional estimate would be an additional assumption. In the general model, M is therefore a separately specified or estimated physical tail absolute-move budget. Given that budget, the symmetric three-state magnitude is

$$m = \frac{M}{p_B + p_U}.$$

4.4 A reference distributional mapping, not an options-implied distribution

The straddle identifies an absolute-move moment under the risk-neutral measure; it does not identify physical probabilities for the three states. To obtain state probabilities, one must specify a physical reference distribution and a threshold a :

$$\begin{aligned} p_B^{\text{ref}} &= \mathbb{P}_{\mathbb{P}}(R^{\text{px}} < -a), \\ p_N^{\text{ref}} &= \mathbb{P}_{\mathbb{P}}(|R^{\text{px}}| \leq a), \\ p_U^{\text{ref}} &= \mathbb{P}_{\mathbb{P}}(R^{\text{px}} > a). \end{aligned}$$

For example, if a centered normal physical reference with scale $s_{\mathbb{P}} > 0$ is imposed, then

$$p_N^{\text{ref}} = 2\Phi\left(\frac{a}{s_{\mathbb{P}}}\right) - 1, \quad p_B^{\text{ref}} = p_U^{\text{ref}} = \frac{1 - p_N^{\text{ref}}}{2}.$$

Equating $s_{\mathbb{P}}$ with a risk-neutral scale inferred from m_{IV} would be an additional physical/risk-neutral equivalence assumption, not a consequence of the straddle. The paper therefore does not assign numerical values to a supposed “options-implied” probability vector. It uses only the explicitly labelled symmetry benchmark $p_B = p_U$ with p_N left symbolic. With symmetric tail magnitudes, that benchmark implies

$$EV_0 = m(p_U - p_B) = 0$$

by identity. A zero produced by this benchmark is not evidence that a predictive model reconstructed market efficiency; no predictive model has been evaluated.

5 Scenario Probabilities

5.1 State labels and probability sources

The thresholds defining B , N , and U must be fixed before outcomes are observed. Once they are fixed, a probability vector can come from one of three distinct sources:

1. the market-implied absolute-move scale together with a separately stated physical probability benchmark;
2. an unconditional historical frequency;
3. a fitted conditional model $\hat{p}_i(x) = \mathbb{P}(Y = i \mid X = x)$.

These sources must not be reported as though they were interchangeable. In particular, the absolute-move moment from options does not supply the physical state probabilities, and any benchmark vector is an assumption rather than a result of the conditional model.

5.2 Ordered probability model

Because $B \prec N \prec U$, one possible parametric specification is an ordered-logit model. This is a modeling choice, not a consequence of the state ordering. Let

$$y^* = x^\top \beta + \varepsilon, \quad \varepsilon \sim \text{Logistic}(0, 1), \quad c_1 < c_2.$$

The unit scale in the logistic error is a normalization for identification, not a calibrated dispersion value. With $\Lambda(z) = 1/(1 + e^{-z})$,

$$\begin{aligned}\mathbb{P}(Y = B \mid x) &= \Lambda(c_1 - x^\top \beta), \\ \mathbb{P}(Y = N \mid x) &= \Lambda(c_2 - x^\top \beta) - \Lambda(c_1 - x^\top \beta), \\ \mathbb{P}(Y = U \mid x) &= 1 - \Lambda(c_2 - x^\top \beta).\end{aligned}$$

The ordering reduces the parameter count and imposes the economically coherent restriction that a feature shifting probability toward U shifts it away from B . An unrestricted multinomial model can be used, but it requires an explicit reference-class constraint; otherwise its latent scores are unidentified under common shifts.

5.3 Separate surprise and reaction channels

Fundamental variables may predict the earnings surprise, while positioning variables may predict the price reaction conditional on that surprise. A conceptual factorization is

$$\mathbb{P}(R^{\text{px}}) = \sum_s \mathbb{P}(R^{\text{px}} \mid s, \text{positioning}) \mathbb{P}(s \mid \text{fundamentals}).$$

This distinction is structural. A strong earnings result need not produce a positive event return if optimistic expectations were already priced. Combining the channels without a stated factorization can turn a reaction effect into a spurious earnings-quality effect.

5.4 Regularization and temporal ordering

If the ordered-logit model is estimated, one possible regularized objective is

$$\hat{\beta} = \arg \min_{\beta} \left[- \sum_{t=1}^T \ln \mathbb{P}(Y_t \mid X_t; \beta) + \eta \|\beta\|_2^2 \right],$$

where $\eta \geq 0$ is selected using time-ordered validation. Standardization constants must also be computed on the training window alone. These are requirements for a future empirical study, not evidence that the model has already been fitted.

6 From Fair Value to Event Price

6.1 The convergence coefficient

Fair value and market price are different variables. A minimal bridge is

$$P_1 = P_0 + \kappa(V_1 - P_0) + \epsilon, \quad \mathbb{E}[\epsilon \mid V_1] = 0.$$

Dividing by P_0 gives

$$R^{\text{px}} = \kappa R^{\text{val}} + \frac{\epsilon}{P_0}, \quad \mathbb{E}[R^{\text{px}}] = \kappa \mathbb{E}[R^{\text{val}}].$$

The coefficient κ is not a convenience parameter. It is the statement about how much of a modeled value gap becomes price movement over the event horizon. The core theory leaves its sign

and magnitude unrestricted. An application may impose a partial-convergence or no-overshoot restriction, but that would be an additional modeling assumption. Treating the entire modeled gap as an immediate event return is the special full-convergence case and is not implied by the valuation equations.

6.2 Why an intercept is required

An empirical version of the bridge should be written across events as

$$R_{j,t}^{\text{px}} = \alpha + \kappa R_{j,t}^{\text{val}} + \epsilon_{j,t}.$$

The intercept is not optional. A common event premium, omitted variable, or other non-valuation component belongs in α rather than being forced into κ . A regression through the origin can therefore attribute a mean event return to the valuation gap and inflate the apparent convergence coefficient.

6.3 Measurement error

The valuation gap is itself estimated. If the measured regressor equals $R^{\text{val}} + u$, where u is mean-zero classical measurement error independent of the true regressor and structural error with variance ς_V^2/P_0^2 , then the usual attenuation expression is

$$\text{plim } \hat{\kappa} = \kappa \frac{\text{Var}(R^{\text{val}})}{\text{Var}(R^{\text{val}}) + \varsigma_V^2/P_0^2}.$$

The direction is toward zero. A future empirical study must therefore report the valuation uncertainty and must not interpret an uncorrected insignificant slope as proof that valuation has no relation to event returns.

6.4 No double counting

There are two related but distinct ways to describe an event return:

1. a scenario-payoff model assigns R_i^{px} directly to reaction states and computes $\sum_i p_i R_i^{\text{px}}$;
2. a valuation model assigns R_i^{val} and maps it into price through $\kappa \sum_i p_i R_i^{\text{val}}$.

If the valuation gap is already used to define the state payoffs, the second quantity is not an additional return to be added on top. A complete empirical model must specify whether the valuation channel changes the state magnitudes, the state probabilities, or a separate component of the price innovation.

7 Expected Value and Closed Forms

7.1 General expected event return

For state-dependent price returns,

$$EV = \mathbb{E}[R^{\text{px}}] = \sum_{i \in \{B, N, U\}} p_i R_i^{\text{px}}.$$

The notation EV means expected value in the statistical sense only. Enterprise value is Θ throughout.

For the symmetric event-payoff model $R^{\text{px}} \in \{-m, 0, +m\}$,

$$\boxed{EV = m(p_U - p_B)}.$$

The directional edge is therefore the probability tilt $p_U - p_B$ multiplied by the event magnitude. A separately imposed symmetric benchmark has zero tilt by construction; the straddle alone does not impose that symmetry.

7.2 Asymmetric event payoffs

Let

$$R^{\text{px}} \in \{-m_B, 0, +m_U\}, \quad \rho = \frac{m_B}{m_U} > 0.$$

The adjusted tail absolute-move budget M supplies the moment equation

$$p_B m_B + p_U m_U = M.$$

Solving with $m_B = \rho m_U$ gives

$$\boxed{m_U = \frac{M}{\rho p_B + p_U}, \quad m_B = \frac{\rho M}{\rho p_B + p_U}}.$$

Expected value is then

$$\boxed{EV = M \frac{p_U - \rho p_B}{\rho p_B + p_U}}.$$

The zero-edge condition is

$$\boxed{EV = 0 \quad \Longleftrightarrow \quad p_U = \rho p_B}.$$

The symmetric condition $p_U = p_B$ is a special case, not a general result.

7.3 Symmetric closed forms

For the symmetric model, write, for a nondegenerate tail and return variance,

$$t = p_U - p_B, \quad q = p_U + p_B, \quad m = \frac{M}{q}.$$

The mean, variance, risk-adjusted ratio, and exact discrete Kelly fraction are

$$\begin{aligned}
EV &= \frac{Mt}{q}, \\
\text{Var}(R^{\text{px}}) &= \left(\frac{M}{q}\right)^2 (q - t^2), \\
\frac{EV}{\sigma_{R^{\text{px}}}} &= \frac{t}{\sqrt{q - t^2}}, \\
f^* &= \frac{t}{M}.
\end{aligned}$$

The risk-adjusted ratio is independent of the absolute move scale. The exact Kelly expression follows by maximizing $\sum_i p_i \ln(1 + f R_i^{\text{px}})$ over the three outcomes; it is not a recommendation to use Kelly sizing when probabilities are uncertain.

7.4 Neutral probability and structural invariance

Holding the tail ratio $p_U : p_B$ fixed, the neutral probability affects the event mean only through the leakage term in M when M is constructed from the normal reference in Section 4.3. If M is separately estimated, the neutral probability does not enter the expected-value formula in this three-state payoff specification. This is a useful diagnostic: a complicated three-state model does not automatically gain decision information from estimating p_N .

8 Risk and Estimation Error

8.1 Downside risk

Variance penalizes positive and negative movement symmetrically. For a threshold τ and power $n \geq 1$, define the lower partial moment

$$\text{LPM}_n(\tau) = \sum_i p_i [\max(\tau - R_i^{\text{px}}, 0)]^n.$$

For the conventional zero-return benchmark $\tau = 0$ (a choice of benchmark, not a theoretical implication), downside deviation is $\sqrt{\text{LPM}_2(0)}$. In the symmetric three-state model,

$$\text{LPM}_1(0) = p_B m, \quad \sqrt{\text{LPM}_2(0)} = m \sqrt{p_B}.$$

The resulting Sortino-like ratio is $t/\sqrt{p_B}$, so it is also degenerate under symmetry. A downside-sensitive statistic becomes informative only after asymmetry is admitted.

8.2 Sampling-error floor

For unconditional multinomial proportions based on independent event observations with a fixed probability vector,

$$\text{Var}(\hat{p}_U - \hat{p}_B) = \frac{p_U(1 - p_U) + p_B(1 - p_B) + 2p_U p_B}{n}.$$

Therefore, in the symmetric model, the corresponding multinomial sampling floor for the expected return is

$$\text{se}_{\text{floor}}(\widehat{EV}) = m \sqrt{\frac{p_U(1 - p_U) + p_B(1 - p_B) + 2p_U p_B}{n}}.$$

This is a lower bound on the uncertainty of a fitted conditional model, not its standard error. A conditional probability estimate also carries coefficient uncertainty, covariance, model-selection uncertainty, uncertainty in m , and possible dependence across event observations. A model that reports a point estimate without an interval is therefore incomplete.

8.3 Scale-free significance

Holding m fixed in the multinomial sampling calculation, both EV and its sampling floor scale linearly in m , so their ratio does not depend on the absolute event magnitude. Changing the straddle-to-tail mapping can change the dollar or percentage estimate without changing the basic significance problem. Directional information is the bottleneck.

8.4 Illustrative sensitivity

Let $(\tilde{p}_B, \tilde{p}_N, \tilde{p}_U)$ denote any hypothetical conditional probability vector with $\tilde{p}_U > \tilde{p}_B$. It is distinct from the symmetric benchmark and is used only to show how a directional tilt enters the formulas. Under symmetry,

$$EV_{\text{sym}} = M \frac{\tilde{p}_U - \tilde{p}_B}{\tilde{p}_B + \tilde{p}_U}.$$

If the downside magnitude is asymmetric, then

$$EV_{\text{asym}} = M \frac{\tilde{p}_U - \rho \tilde{p}_B}{\rho \tilde{p}_B + \tilde{p}_U}.$$

The edge is exactly zero at $\rho = \tilde{p}_U / \tilde{p}_B$ and changes sign above that value. No numerical probability vector or physical-move haircut is needed to establish this result.

9 Theoretical Implications and Empirical Program

9.1 What the theory predicts

The framework produces testable implications without claiming that they have already been tested:

1. A fitted probability model must beat both an unconditional historical base rate and a pre-specified physical benchmark on proper out-of-sample scoring rules if it is to claim incremental probability information.
2. The valuation layer should be evaluated through an intercept-inclusive estimate of κ , with measurement error in the value gap reported.
3. The downside-to-upside ratio ρ should be estimated before a directional expected value is interpreted, because it can reverse the sign of the result.
4. Any economic evaluation should use event returns and appropriate transaction-cost, liquidity, and alternative-investment benchmarks rather than treating zero return as the only alternative.

9.2 A falsification protocol

A future empirical study should use time-ordered training and test periods, walk-forward refitting, and probability calibration metrics such as Brier score, log loss, and expected calibration error. For probabilities $\hat{p}_{i,t}$ and realized state Y_t ,

$$\text{Brier} = \frac{1}{T} \sum_{t=1}^T \sum_i (\hat{p}_{i,t} - \mathbf{1}\{Y_t = i\})^2, \quad \text{LogLoss} = -\frac{1}{T} \sum_{t=1}^T \ln \hat{p}_{Y_t,t}.$$

The economic test is separate from the probability test. A well-calibrated probability model need not generate a profitable strategy after costs, and a profitable strategy need not identify which part of the return came from valuation, positioning, or a broad announcement premium.

9.3 Position in the literature

The framework is consistent with the literature’s distinction between event information, announcement returns, and post-earnings drift. Earnings announcements have long been treated as information events [3, 1, 5]. Cross-sectional announcement premia and their limits to arbitrage have been documented in diversified portfolios [6, 7, 9, 2]. Post-earnings drift is a separate phenomenon because the surprise is known after the event [4, 8].

Those findings motivate the model’s caution; they do not validate a particular single-name forecast or any parameter used in the illustrations.

10 Limitations and Boundary Conditions

1. The three-state approximation discards fat tails and path dependence. This is a deliberate low-dimensional representation, not a claim that the true event distribution has three points.
2. h , the physical adjustment from the risk-neutral straddle moment, is not identified here.
3. ρ is not estimated. Since it determines the zero-edge condition, no asymmetric expected-value conclusion is empirical in this paper.
4. κ is not estimated. A fair-value gap does not become an event return without an empirically supported price-convergence relation.
5. The surprise variables require forecasts of actual earnings, revenue, guidance, and persistence. If those forecasts add no information beyond consensus, the fundamental branch contributes no incremental edge.
6. Implied-volatility crush, liquidity, trading halts, guidance withdrawal, taxes, and transaction costs require separate models when an economic application is attempted.
7. The straddle identifies an absolute-move moment, not physical state probabilities. Any numerical probability benchmark is an explicit distributional assumption and should not be described as a forecast.

11 Conclusion

The main theoretical result is a negative one in the useful scientific sense: a pre-earnings expected-value claim is underdetermined when it reports only a fair-value gap and a symmetric implied move. The straddle supplies an absolute move moment, not a tail magnitude. A neutral band absorbs part of that moment. Directional value depends on a probability tilt, while the sign under realistic asymmetry depends on the ratio of downside to upside magnitudes. A valuation gap reaches price only through a convergence coefficient, and the uncertainty of the probability tilt must be quantified relative to the event counts available for a single name.

The framework therefore defines a disciplined research object rather than a validated trading system. Its null is explicit, its parameters are separated, its assumptions are visible, and its empirical requirements are stated. A future study may estimate the parameters and test the implications. Until then, the theory's appropriate conclusion is not that an edge exists, but that any claimed edge must survive the magnitude, asymmetry, convergence, and uncertainty tests derived here.

A Secondary Decision Quantities

This appendix records quantities that are useful for an eventual application but are not needed for the theoretical contribution.

A.1 Mean-variance utility

For an action a with event return R_a^{px} , a local mean-variance approximation to a CRRA objective, after normalizing by a reference wealth level, is

$$U(a) = \mathbb{E}[R_a^{\text{px}}] - \frac{\gamma}{2} \text{Var}(R_a^{\text{px}}) - C(a),$$

where $\gamma > 0$ is an unspecified relative-risk-aversion parameter and $C(a)$ collects trading, tax, liquidity, and other action-specific costs. The comparison is against the best alternative, not automatically against zero. For the costless analytical benchmark with the alternative normalized to zero, the break-even risk-aversion coefficient in the symmetric model is

$$\gamma^* = \frac{2tq}{M(q - t^2)}.$$

This expression is a sensitivity result. It is not a recommendation to infer an investor's risk aversion from a single event.

A.2 Kelly sizing

Under known probabilities, no transaction costs, and no position or leverage constraints, the exact growth-optimal fraction in the symmetric three-state model, over the feasible domain $1 + fR_i^{\text{px}} > 0$, is

$$f^* = \frac{t}{M}.$$

The exact expression does not account for parameter uncertainty, costs, constraints, or model risk. Any practical use would therefore require an externally specified decision rule for shrinking or otherwise restricting the fraction.

B Illustrative Calculation

The following calculation is included only to make the identities concrete. It is not a stock forecast and uses no observed event data.

Let a be a pre-registered neutral threshold and let a hypothetical centered normal physical reference distribution be $\mathcal{N}(0, s_{\mathbb{P}}^2)$, with $s_{\mathbb{P}} > 0$. Then

$$p_N^{\text{ref}} = 2\Phi\left(\frac{a}{s_{\mathbb{P}}}\right) - 1, \quad p_B^{\text{ref}} = p_U^{\text{ref}} = \frac{1 - p_N^{\text{ref}}}{2}.$$

For any hypothetical conditional vector $(\tilde{p}_B, \tilde{p}_N, \tilde{p}_U)$ with nonnegative components summing to one, define

$$\tilde{q} = \tilde{p}_B + \tilde{p}_U, \quad \tilde{t} = \tilde{p}_U - \tilde{p}_B, \quad M > 0.$$

Here M is a separately specified physical tail absolute-move budget; it is not inferred from $\tilde{p}_N$ unless the additional normal-reference mapping and physical-scale assumptions are imposed.

The symmetric and asymmetric expected values are respectively

$$EV_{\text{sym}} = M \frac{\tilde{t}}{\tilde{q}}, \quad EV_{\text{asym}} = M \frac{\tilde{p}_U - \rho \tilde{p}_B}{\rho \tilde{p}_B + \tilde{p}_U}.$$

When $\tilde{p}_B > 0$, the asymmetric edge changes sign at $\rho = \tilde{p}_U / \tilde{p}_B$. This symbolic result is the point of the illustration; no arbitrary numerical probability vector is needed.

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